Andrea Della Vecchia, PhD

Current Position	PostDoc - EPFL
Work Permit	Switzerland; EU citizen
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Publications	Google Scholar
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Summary: Researcher with a PhD in Applied Maths and a strong interest in applying ML and statistical methods to quantitative finance. International research experience with publications and presentations in top-tier venues including NeurIPS, ICML, JMLR, and AISTATS. Expertise in developing mathematically grounded ML algorithms with rigorous statistical guarantees and scalable and efficient implementations. Research focus in ML and deep learning for time series forecasting, asset pricing, portfolio optimization, and alpha generation.

Work Experience

- **Quantitative Research Collaborator, MFM Mirante Fund Management** Lausanne, CH | June 2026 – Present
  Academic–industry collaboration on quantitative research and systematic investment strategies.
  – Developing predictive models and data-driven methodologies on proprietary financial datasets.
  ⋆ Gaussian Process models for return forecasting, with applications to systematic portfolio construction.
  ⋆ Deep learning–based models for joint estimation of conditional expected returns and covariance structures through systematic and idiosyncratic components.
- **PostDoc in ML & Quant Finance, Swiss Finance Institute @ EPFL** Lausanne, CH | Sept. 2024 – Present
  Supervisor: Prof. Damir Filipović (Swissquote Chair in Quant Finance at EPFL)
  – Kernel-based methods for dynamic programming in stochastic control and American option pricing (ICML26).
  ⋆ Statistical error-propagation analysis and development of fast solvers for high-dimensional pricing.
  – Density ratio estimation for large-scale conditional distribution learning and hypothesis testing (JMLR)
- **Research Collaborator, Leonardo Labs (AI Research Center of Leonardo S.p.A.)** Genoa, IT | 2023 – 2024
  Academic–industry collaboration on applied ML research.
  – Robust and scalable learning for non i.i.d. data: statistical guarantees under covariate shift (NeurIPS25).
  ⋆ Nyström–KRR implementation with shift-aware efficiency; extensive simulations and real data testing.
- **PostDoc in ML & Statistics, IIT@MIT Lab, Italian Institute of Technology** Genoa, IT | 2023 – 2024
  Efficiency in ML: generalization guarantees for random projection techniques, i.e. Nyström method, for convex losses (JMLR); scalable kernel instrumental variables; kernel methods vs deep learning in weather prediction.
- **Spring Insight Program – Quantitative Research, G-Research** London, UK | 2022
  Hands-on tutorials on factor modeling, signal generation, and backtesting frameworks in hedge funds.
- **Visiting PhD, LAILA - Laboratory of Artificial Intelligence and Learning Algorithms, UniMi** Milan, IT | 2022
  Regret bounds for adaptive multitask online learning and implementation of the corresponding algorithm *Adatask*.
- **Research Intern, Max Planck Institute for Intelligent Systems (MPI-IS)** Tübingen, DE | 2019
  Dynamic hierarchical ranking (Physical Review E), Bayesian models, variational inference.

Education

PhD in Applied Maths and Machine Learning, University of Genoa Genoa, IT | 2019 – 2022
Thesis: *"Optimal and Efficient Learning In Classification"* | Advisors: Lorenzo Rosasco, Ernesto De Vito
- Research: statistical learning theory, multitask learning, online learning, optimization.

MSc in Data Science & ML, University of Padua Padua, IT | 2017 – 2019
GPA: 29/30 | Final grade: 110/110 with honours.
Relevant coursework: *Business, Economic and Financial Data; Stochastic Methods; Game Theory*

First year MSc in Mathematics, University of Padua Padua, IT | 2016 – 2017
Completion of the 1st year before transitioning to the MSc program in Data Science.
Relevant coursework: *Stochastic Methods for Finance; Stochastic Analysis.*

BSc in Physics, University of Padua Padua, IT | 2013 – 2016
Thesis in Statistical Physics: *"Diffusion and Brownian Motion".*

Supervision & Teaching

- **Mentorship and co-supervision of student research projects** 2023 - Present
  2 PhD students (PhD in Appl. Maths), 1 Master's student (MSc Financial Eng.) and 1 Bachelor's student (BSc Eng.) Project/Thesis topics: deep learning for return and risk forecasting in portfolio optimization; unsupervised learning methods for bond rating refinement; efficient algorithms for stochastic optimal control problems with streaming data; statistical guarantees in ML under distribution shift.
- **Teaching Assistant, MSc in Financial Engineering (EPFL) and MSc in Engineering (UniGe)** 2020 - Present
  Probability & Stochastic Calculus, *Convex Optimization*, *Financial Applications of Blockchains and Distributed Ledgers*, *Mathematical Analysis*, *Applied Mathematics*.

Awards & Honors

– **NeurIPS 2025 Spotlight Presentation** (top 3% of submitted papers).
– **Italian Mathematical Olympiad**: Silver and Bronze Medals (individual competition); finalist (team competition).
– **Italian Physics Olympiad**: finalist (individual competition).
– **G-Research Spring Insight Program in Quant Finance**: selected after technical screening among top ~ 40 applicants.
– **Academic Scholarships**: full PhD scholarship (University of Genoa); MSc scholarship awarded to top two students in Data Science (University of Padua).

Selected First-Author Publications

1. *"Post-Processing Guarantees for Classification under Linear-Fractional Performance Metrics"*, A. Della Vecchia. **NeurIPS 26**, Sydney, 2026.
2. *"Error Propagation in Dynamic Programming: From Stochastic Control to Option Pricing"*, A. Della Vecchia, D. Filipović. **ICML 26**, Seoul, 2026.
3. *"Kernel Density Machine"*, A. Della Vecchia, D. Filipović, P. Schneider. **Under review at JMLR**, 2026.
4. *"Computational Efficiency under Covariate Shift in Kernel Ridge Regression"*, A. Della Vecchia, A. Mavakala, E. De Vito, L. Rosasco. **NeurIPS 25 (Spotlight)**, San Diego, 2025.
5. *"The Nyström method for convex loss functions"*, A. Della Vecchia, E. De Vito, J. Mourtada, L. Rosasco. **Journal of Machine Learning Research (JMLR)**, 2024.
6. *"A model for efficient dynamical ranking in networks"*, A. Della Vecchia, K. Neocosmos, D.B. Larremore, C. Moore, C. De Bacco. **Physical Review E**, 2024.
7. *"Regularized ERM on random subspaces"*, A. Della Vecchia, J. Mourtada, E. De Vito, L. Rosasco. **AISTATS**, 2021.

Full list available on Google Scholar.

Technical Skills

– **Quantitative research**: statistical modelling, stochastic optimal control, factor models, backtesting, time-series forecasting, option pricing.
– **Machine Learning**: statistical learning, deep learning, online learning, kernel methods, dynamic programming, convex optimization, distribution shift, ensemble and unsupervised methods.
– **Tools**: `Python`, `C/C++`, `SQL`, `Bash`, `R`, `PyTorch`, `scikit-learn`, `statsmodels`, `NumPy`, `SciPy`, `pandas`, `Git`, `Linux`.

Languages

– **English**: full professional. **Italian**: native.

Other Interests

– Competitive 11-a-side soccer (regional champion, 2011).
– Cinema, hiking, tennis, poker, value investing.
– Volunteering: math tutor and students mentor; summer camp organizer for primary school children.